

Jan Škvrna

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Research Interests

3D Object Detection • Weakly Supervised & Self-Supervised Learning • 3D Reconstruction • Autonomous Driving

Education

PhD in Informatics

Feb 2024 – Present

Czech Technical University in Prague

Visual Recognition Group (Prof. Jiří Matas), supervised by Dr. Lukáš Neumann. Research on weakly-supervised 3D object detection for autonomous driving.

MSc in Cybernetics and Robotics — with Honours

Sep 2021 – Feb 2024

Czech Technical University in Prague

Thesis: Weakly-supervised 3D object detection using temporally consistent 2D cues. **Dean's Award** for exceptional thesis. Published at ECCV 2024.

BSc in Cybernetics and Robotics

Sep 2018 – Jun 2021

Czech Technical University in Prague

Erasmus exchange at University of Southern Denmark (Sep 2022 – Feb 2023).

Publications

J. Škvrna, L. Neumann. MonoSOWA: Scalable Monocular 3D Object Detector Without Human Annotations. *International Conference on Computer Vision (ICCV)*, 2025. [🔗 CVF](#)

J. Škvrna, L. Neumann. TCC-Det: Temporarily Consistent Cues for Weakly-Supervised 3D Detection. *European Conference on Computer Vision (ECCV)*, 2024. [🔗 Springer](#)

Awards & Honors

Winning Solution — Structured Semantic 3D Reconstruction (S23DR) Challenge 2025
Part of the 2nd Urban Scene Modeling Workshop at **CVPR 2025**, Nashville. \$25k USD prize pool.

Dean's Award for Exceptional Master's Thesis 2024
Czech Technical University in Prague.

6th Best Master's Thesis on Industry 4.0 in Czech Republic 2024
[Werner von Siemens Award](#).

Industry Experience

Junior Embedded Software Developer

Jun 2020 – Jun 2023

Poll s.r.o.

Developed real-time firmware in C/C++ for ARM MCUs (HVAC, high-voltage DC/DC converters). Contributed to FPGA VHDL implementations and Java-based diagnostics.

Technical Skills

Languages: Python, C/C++, LaTeX

ML / DL: PyTorch, NumPy, OpenCV, Open3D, PyTorch3D, Faiss

Tools: Git, Slurm (large-scale training), Linux

Academic Service

Reviewing: ICCV 2025

Invited Talk: Computer Vision Winter Workshop, Graz, Austria (Feb 2025) — *MonoSOWA*

Summer Schools: International Computer Vision Summer School (ICVSS), Sicily (2025); Vision and Sports, Prague (2024)